BUSINESS SALES PERFORMANCE ANALYSIS & DECISION REPORT

Business Analyst Internship Program — Task 1 Deliverable
Author: Josy Philip
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1. Executive Summary

This report presents a comprehensive sales performance analysis conducted for management to evaluate revenue generation, product profitability, regional dynamics, and seasonal demand patterns. Utilizing transaction-level sales data, the analysis explores five core business hypotheses, measures key performance indicators (KPIs), and outlines actionable, data-driven recommendations to optimize profitability and strategic resource allocation

Key high-level metrics established across 1,194 analyzed transactions

- **Total Sales Revenue:** ₹6,182,639
- **Total Net Profit:** ₹1,610,697
- **Overall Profit Margin:** 26.05%
- **Average Order Value (AOV):** ₹5,178.09
- **Total Quantity Sold:** 1,194 units

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2. Business Understanding & Objectives

Management sought to answer critical operational questions regarding sales drivers, product margins, geographic performance, and seasonal volatility

1. Which products and categories generate the highest sales and profits
2. How do regional markets compare in revenue and profitabilityAre there discernible seasonal buying patterns throughout the year
3. What strategic recommendations can improve overall organizational performance

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3. Data Cleaning & Preparation Methodology

To ensure high data integrity, rigorous cleansing and standardization procedures were executed in Excel:

- **Text Standardization:** Utilized `TRIM()` and `PROPER()` to eliminate trailing spaces and standardize categorical fields such as *Category*, *Sub-Category*, *Customer Name*, *State*, *City*, and *Payment Mode*.
- **Data Type Enforcement:** Converted numeric fields (*Amount*, *Profit*, *Quantity*) using `VALUE()` and standardized order dates using `DATEVALUE()`.
- **Missing Value Handling:** Verified zero missing values across critical business dimensions using `COUNTBLANK()`.
- **Duplicate Detection:** Confirmed transaction validity via `COUNTIF()`, retaining recurring Order IDs intentionally as single orders encompass multiple product line-items.
- **Analytical Helper Columns Created:**
  - *Year:* `=YEAR(H2)`
  - *Month Number:* `=MONTH(H2)`
  - *Month Name:* `=TEXT(H2,"MMMM")`
  - *Quarter:* `="Q"&ROUNDUP(MONTH(H2)/3,0)`
  - *Profit Margin %:* `=IF(B2=0,0,C2/B2)`
  - *Order Size Segmentation:* `=IF(B2<3000,"Low",IF(B2<7000,"Medium","High"))`

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4. Exploratory Data Analysis (EDA) Summary

- **Category Distribution:** Furniture recorded the highest transaction count (407 records), closely followed by Office Supplies and Electronics.
- **Payment Preferences:** Debit Cards (260 transactions) and Credit Cards emerged as the most preferred customer payment methods.
- **Geographic Hotspots:** California led in total state transactions, followed closely by Florida. At the municipal level, Buffalo (90 records) and San Francisco recorded the highest order densities.
- **Transaction Characteristics:** Order amounts exhibited a balanced spread ranging from ₹508 to ₹9,992, with an average transaction value of ₹5,178.09 and an average profit of ₹1,348.99.

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5. Hypothesis Testing & Findings

Hypothesis 1: Product Concentration Effect

- **Hypothesis:** A small number of products generate the majority of sales revenue (80/20 rule).
- **Finding:** ❌ **REJECTED**
- **Evidence:** The top 3 sub-categories (*Markers*, *Tables*, *Sofas*) contribute 29.46% (₹1,821,419) of total sales, while the top 5 contribute 47.76% (₹2,952,870). Revenue is evenly distributed across a diverse product portfolio.
- **Business Insight:** The business exhibits low product concentration risk. Maintaining a diversified inventory prevents over-reliance on a narrow product line.

Hypothesis 2: High Sales Does Not Always Mean High Profit

- **Hypothesis:** Products generating the highest sales revenue do not necessarily deliver the highest profitability.
- **Finding:** ✅ **SUPPORTED**
- **Evidence:**
    - *Phones* and *Pens* generate robust sales revenue (₹503,055 and ₹552,269 respectively) but suffer from lower profit margins of 22.58% and 23.51%.
    - Conversely, *Bookcases* and *Paper* achieve superior profit margins of 28.56% and 28.53% despite lower gross sales volumes.
- **Business Insight:** Management must shift from a revenue-only evaluation model to a margin-conscious strategy, optimizing pricing and discounting structures for high-volume, low-margin lines.

Hypothesis 3: Regional Performance Differences

- **Hypothesis:** Sales and profitability vary significantly across geographic regions.
- **Finding:** ✅ **SUPPORTED**
- **Evidence:** New York generated the highest total revenue (₹1.13M), whereas Ohio recorded the lowest sales (₹0.88M)—a regional spread of ₹245,280. Furthermore, Florida demonstrated superior efficiency with the highest profit margin (28.29%) and total profit (₹308,706), compared to Ohio's lagging margin of 24.47%.
- **Business Insight:** Geographic performance is non-uniform. Capital and marketing spend should be doubled down in high-yielding states like Florida and New York, while investigating root causes for Ohio's underperformance.

Hypothesis 4: Seasonal Sales Patterns Exist

- **Hypothesis:** Sales fluctuate noticeably throughout the year due to seasonal demand.
- **Finding:** ✅ **SUPPORTED**

- **Evidence:** December recorded the peak sales month at ₹655,378, followed closely by October (₹632,521). Conversely, January experienced the seasonal trough at ₹392,678—representing a 67% surge in December over January.
- **Business Insight:** Strong cyclical peaks require proactive inventory stocking and pre-season marketing campaigns ahead of October and December, coupled with targeted promotional pushes during slow months like January and September.

Hypothesis 5: One Category Dominates Revenue

- **Hypothesis:** One product category contributes a significantly larger share of total revenue than others.
- **Finding:** ❌ **REJECTED**
- **Evidence:** Revenue distribution across categories is remarkably balanced: Office Supplies (33.80% / ₹2,089,510), Electronics (33.23%), and Furniture (32.97% / ₹2,038,673). Notably, Furniture achieved the highest category profit margin at 26.51%.
- **Business Insight:** The organization maintains a highly resilient, multi-pillar revenue structure with minimal dependency risk.

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6. Strategic Business Recommendations

1. **Prioritize High-Margin Product Growth:** Allocate inventory and promotional budget toward top-margin lines such as *Bookcases* (28.56% margin), *Paper* (28.53%), and *Chairs* (28.45%).
2. **Revise Pricing on High-Volume, Low-Margin Items:** Conduct cost-to-serve and discounting audits for *Phones* and *Pens* to elevate their profitability closer to company benchmarks.
3. **Optimize Regional Resource Allocation:** Double down on marketing in Florida and New York while conducting a strategic review of operational bottlenecks in Ohio.
4. **Synchronize Supply Chain with Seasonality:** Build inventory buffers ahead of Q4 peaks (October and December) and deploy aggressive discounting campaigns during the January slow period to smooth revenue volatility.

Report compiled successfully for academic and corporate evaluation.